

ALFAAH65892

## N-Isopropylmethylamine

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** N-异丙基甲胺  
N-Isopropylmethylamine

**Cat No. :** H65892  
**Synonyms** N-Methylisopropylamine  
**CAS No** 4747-21-1  
**Molecular Formula** C4 H11 N

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
No information available

**Odor**  
No information available

#### Emergency Overview

Toxic if swallowed. Causes severe skin burns and eye damage. Highly flammable liquid and vapor. Toxic if inhaled. Air sensitive.

#### Classification of the substance or mixture

|                                    |              |
|------------------------------------|--------------|
| Flammable liquids.                 | Category 2   |
| Acute Oral Toxicity                | Category 3   |
| Acute Inhalation Toxicity - Vapors | Category 3   |
| Skin Corrosion/Irritation          | Category 1 A |
| Serious Eye Damage/Eye Irritation  | Category 1   |

#### Label Elements



## N-Isopropylmethylamine

## Signal Word

Danger

## Hazard Statements

H225 - Highly flammable liquid and vapor  
H314 - Causes severe skin burns and eye damage  
H301 + H331 - Toxic if swallowed or if inhaled

## Precautionary Statements

## Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

## Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Highly flammable. Vapors may cause flash fire or explosion.

## Health Hazards

Toxic if swallowed. Corrosive. Causes skin and eye burns. Toxic if inhaled.

## Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its volatility. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces.

This product does not contain any known or suspected endocrine disruptors. Toxic to terrestrial vertebrates.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component              | CAS No    | Weight % |
|------------------------|-----------|----------|
| N-Methylisopropylamine | 4747-21-1 | >95      |

## SECTION 4. FIRST AID MEASURES

## General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

## Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

## Skin Contact

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Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms and effects**

Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Water.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE**

## N-Isopropylmethylamine

**Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Flammables area. Keep away from heat, sparks and flame. Store under an inert atmosphere.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Exposure Controls****Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment**

**Eye Protection** Goggles (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Nitrile rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Neoprene       | recommendations   |                 |             |                       |
| Natural rubber |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection** No protective equipment is needed under normal use conditions.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Small scale/Laboratory use** Maintain adequate ventilation

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**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                |                             |                                             |
|------------------------------------------------|-----------------------------|---------------------------------------------|
| <b>Appearance</b>                              |                             |                                             |
| <b>Physical State</b>                          | Liquid                      |                                             |
| <b>Odor</b>                                    | No information available    |                                             |
| <b>Odor Threshold</b>                          | No data available           |                                             |
| <b>pH</b>                                      | No information available    |                                             |
| <b>Melting Point/Range</b>                     | No data available           |                                             |
| <b>Softening Point</b>                         | No data available           |                                             |
| <b>Boiling Point/Range</b>                     | 50 - 53 °C / 122 - 127.4 °F |                                             |
| <b>Flash Point</b>                             | -32 °C / -25.6 °F           | <b>Method -</b> No information available    |
| <b>Evaporation Rate</b>                        | No data available           |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable              | Liquid                                      |
| <b>Explosion Limits</b>                        | No data available           |                                             |
| <b>Vapor Pressure</b>                          | 286 hPa (25°C)              |                                             |
| <b>Vapor Density</b>                           | No data available           | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.702                       |                                             |
| <b>Bulk Density</b>                            | Not applicable              | Liquid                                      |
| <b>Water Solubility</b>                        | No information available    |                                             |
| <b>Solubility in other solvents</b>            | No information available    |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                             |                                             |
| <b>Component</b>                               | <b>log Pow</b>              |                                             |
| N-Methylisopropylamine                         | 0.52                        |                                             |
| <b>Autoignition Temperature</b>                | No data available           |                                             |
| <b>Decomposition Temperature</b>               | No data available           |                                             |
| <b>Viscosity</b>                               | No data available           |                                             |
| <b>Explosive Properties</b>                    |                             | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available    |                                             |
| <b>Molecular Formula</b>                       | C4 H11 N                    |                                             |
| <b>Molecular Weight</b>                        | 73.14                       |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                                         |                                                                                                                        |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Air sensitive.                                                                                                         |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                          |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                              |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. Exposure to air. |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Strong acids. Acid anhydrides. Acid chlorides.                                                |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Nitrogen oxides (NO <sub>x</sub> ).                           |

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

# SAFETY DATA SHEET

## N-Isopropylmethylamine

**(a) acute toxicity;**

| Component              | LD50 Oral | LD50 Dermal | LC50 Inhalation                                              |
|------------------------|-----------|-------------|--------------------------------------------------------------|
| N-Methylisopropylamine |           |             | LC50 = 9.39 mg/L ( Rat ) 4 h<br>LC50 = 7.11 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** Category 1 A**(c) serious eye damage/irritation;** Category 1**(d) respiratory or skin sensitization;**

|             |                   |
|-------------|-------------------|
| Respiratory | No data available |
| Skin        | No data available |

**(e) germ cell mutagenicity;** No data available**(f) carcinogenicity;** No data available  
There are no known carcinogenic chemicals in this product**(g) reproductive toxicity;** No data available**(h) STOT-single exposure;** No data available**(i) STOT-repeated exposure;** No data available

|               |             |
|---------------|-------------|
| Target Organs | None known. |
|---------------|-------------|

**(j) aspiration hazard;** No data available**Other Adverse Effects** The toxicological properties have not been fully investigated.**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation**SECTION 12. ECOLOGICAL INFORMATION****Ecotoxicity effects** Do not empty into drains. .**Persistence and Degradability** No information available  
**Persistence** Persistence is unlikely, based on information available.**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component              | log Pow | Bioconcentration factor (BCF) |
|------------------------|---------|-------------------------------|
| N-Methylisopropylamine | 0.52    | No data available             |

**Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all

## N-Isopropylmethylamine

surfaces Will likely be mobile in the environment due to its volatility Disperses rapidly in air

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

UN-No UN2734  
Proper Shipping Name AMINES, LIQUID, CORROSIVE, FLAMMABLE, N.O.S.  
Technical Shipping Name N-Methylisopropylamine  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group I

**IMDG/IMO**

UN-No UN2734  
Proper Shipping Name AMINES, LIQUID, CORROSIVE, FLAMMABLE, N.O.S.  
Technical Shipping Name N-Methylisopropylamine  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group I

**IATA**

UN-No UN2734  
Proper Shipping Name AMINES, LIQUID, CORROSIVE, FLAMMABLE, N.O.S.  
Technical Shipping Name N-Methylisopropylamine  
Hazard Class 8  
Subsidiary Hazard Class 3  
Packing Group I

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The | List of | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-----------|-----|---------|------|-------|--------|------|-----|-------|------|------|------|------|
|-----------|-----|---------|------|-------|--------|------|-----|-------|------|------|------|------|

**SAFETY DATA SHEET****N-Isopropylmethylamine**

|                        |                                                             |                                          |   |   |           |   |   |   |   |   |   |   |
|------------------------|-------------------------------------------------------------|------------------------------------------|---|---|-----------|---|---|---|---|---|---|---|
|                        | Inventory of<br>Hazardous<br>Chemicals<br>(2015<br>Edition) | dangerous<br>goods GB<br>12268 -<br>2012 |   |   |           |   |   |   |   |   |   |   |
| N-Methylisopropylamine | -                                                           | -                                        | X | X | 225-266-7 | - | - | - | - | X | - | - |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 06-Dec-2010  
**Revision Date** 29-Apr-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage,



**SAFETY DATA SHEET**

N-Isopropylmethylamine

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transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**